



AWS Certified Solutions Architect - Associate (SAA-C03)



Any Domain - Any Workload - One OS

Course Description

AWS Certified Solutions Architect - Associate showcases knowledge and skills in AWS technology, across a wide range of AWS services. The focus of this certification is on the design of cost and performance optimized solutions, demonstrating a strong understanding of the AWS Well-Architected Framework. This certification can enhance the career profile and earnings of certified individuals and increase your credibility and confidence in stakeholder and customer interactions.

Audience

- Cloud Native Developers
- System Administrator
- Network Engineers
- System Engineers
- Cloud Engineers
- Site Reliability Engineers
- SME

Prerequisite

- Experience in AWS technology.
- Strong on-premises IT experience and understanding of mapping on-premises to cloud.
- Experience working in other cloud services.

S. NO	DESCRIPTION	PAGE NO
CHAPTER 1	Introduction to Cloud and AWS	
	Cloud Service Model	
	AWS Introduction	
	AWS Global Infrastructure	
	AWS Region Availability Zones	
	Replication of Data Between Regions	
	AWS Services	
	Create AWS Free Tier Account	
	Billing Alarms	
	AWS Service Limits	
	AWS Ec2 Machine	
	LAB	
	Create AWS Free Trial Account	
	Create Billing Alarms & Service Limits	
	Create & Connect to Windows Ec2 Machine	
	Host Website on Windows Ec2 Instance using IIS	
	Create & Connect to Linux Ec2 Machine	
	Install & Configure AWS CLI	



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CHAPTER 2	Security Management AWS	
	Identity and Access Management IAM	
	Best Practices of IAM	
	Creating API Keys for Accessing AWS Services	
	Key Management System (KMS)	
	Guardduty Detection Control	
	Infrastructure Protection VIA WAF & Shield	
	Data Protection VIA KMS and Cloud HSM	
	Lab	
	Working with AWS IAM	
	Enable Multi-Factor Authentication	
	AWS KMS Create & Use	
	IAM Power User	
CHAPTER 3	Object Storage Options	
	Traditions VS Cloud Storage	
	Different Storage Options on AWS	
	Difference Between S3, EBS and EFS	
	Cross-Region Replication (CRR) and its Use Case	
	Amazon S3 Transfer Acceleration	
	Cloud Front and Snowball	
	Storage Gateway	
	Glacier Vault Lock & S3 Object Lock	
	Lab	
	Create S3 Bucket, Upload & Access Files and Host Website	
	S3 Cross Region Replication	
	S3 Lifecycle Management on S3 Bucket	
CHAPTER 4	Designing Computing Environment	
	Compute Virtualization	
	Amazon Elastic Compute Cloud (Ec2)	
	Amazon Machine Image (AMI)	
	Hardware Tenancy Shared VS Dedicated	
	Elastic Network Interface (ENI)	
	Instance Store	
	Elastic Block Store (EBS)	
	Solid State Drive & Hard Disk Drive	
	Elastic File System (EFS)	
	Lab	
	Create and Manage EBS Volumes & Snapshots	
	Attach & Mount EBS Volume to Linux Ec2 Instance	
	Attach & Mount EBS Volume to Ec2 Windows Instance	
	Create Windows File Systems Using Amazon FSX	
	Create & Mount Elastic File System (EFS) on Ec2 Instances	
	SSH into Ec2 Linux Instance VIA Username & Password	



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CHAPTER 5	Load Balancer, Auto Scaling & Route 53	
	Elastic Load Balancer	
	Application V/S Network V/S Gateway	
	Load Balancer	
	Load Balancer Troubleshoot	
	Auto Scaling and Lifecycle	
	Route 53	
	Routing Policies	
	Lab	
	Network Load Balancer	
	Configure A Load Balancer and Autoscaling on Ec2 Instances	
	Amazon Ec2 Auto Scaling with Ec2 Spot Instances	
	Block Web Traffic with WAF in AWS	
	Register A Domain Name for Free	
	Map DNS Using Route 53	
	Create Launch Template for Auto Scaling Group	
CHAPTER 6	Networking & Monitoring Services	
	Traditional and Virtual Network	
	Virtual Private Cloud (VPC)	
	CIDR-Classless Inter-Domain Routing	
	Subnet, Internet Gateway and Route Tables	
	Control List, NACL and Security Group	
	NAT Devices, Gateway and Instance	
	VPC Peering and its Working	
	VPN and Direct Connctet	
	AWS Cloud Watch	
	Lesson 10: Cloud Trail	
	Lesson 11: Trusted Advisor and its Working	
	Lab	
	Create Custom Virtual Private Cloud	
	Work with VPC Peering Connection	
	AWS Elastic IP	
	Configure Amazon Cloudwatch to Notify Change in Ec2 CPU Utilization	
	Enable Cloud Trail and Store Logs in S3	
	Setting up AWS Config to Assess Audit & Evaluate AWS Resources Lab 7: Establish A Client-Side VPN	
CHAPTER 7	Database Server & Analytics	
	Database and Types of AWS Database Services	
	Amazon RDS and Read Replica	
	IAM in Amazon RDS	
	SQL and NOSQL Databases	
	Amazon Dynamodb	
	Amazon ElastiCache, Memcached & Redis	
	Amazon Redshift and Architecture	
	Amazon Aurora	



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CHAPTER 7	Database Server & Analytics	
	Lab	
	Create & Query with Amazon Dynamodb	
	Create A Redis Cache & Connect it to Ec2 Instance	
	Visualize Web Traffic Using Kinesis Data Streams	
	Amazon Athena	
	Amazon Aurora	
CHAPTER 8	Application & Messaging Services	
	Simple Email Service (SES)	
	Simple Notification Service (SNS)	
	SNS Mobile Push	
	Simple Queue Service (SQS)	
	Long Polling, Visibility Timeout and Dead Letter Queues	
	Simple Workflow Service (SWF)	
	Serverless Computing Lambda	
	Lab	
	Send an E-Mail Through AWS SES	
	Event-Driven Architectures using AWS Lambda, SES, SNS and SQS	
CHAPTER 9	Configuration Management & Automation	
	Infrastructure as a Code	
	Cloud Formation and its Components	
	AWS OPS Works Services	
	Elastic Beanstalk	
	Beanstalk VS, OPS Works VS Cloud Formation	
	Lab	
	Create and Update Stacks Using Cloud Formation	
	Deploy a Web Application Using Elastic Beanstalk	
CHAPTER 10	Architectures in AWS	
	AWS Well. Architected Framwork	
	Pillars of AWS Well-Architected	
	Highly Available And/Or Fault-Tolerant Architectures	
	Resilient Storage	
	Decoupling Mechanism Using AWS	
	Multi-Tier Architecture Solution	
	Disaster Recover DR Implementation Plans	

- Note: 1. The list of Lab Exercises a student should do during their lab sessions is given above
2. Students are requested to maintain all completed lab exercises with the outputs in the Lab Observation Records